



Machine Vision Engineering Handbook

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1. Machine Vision Engineering Handbook

Welcome

The Machine Vision Engineering Handbook is the principal technical knowledge resource of the Machine Vision Academy.

Warning

*In order to make content available as soon as it is developed and reviewed, this handbook has been published with structural placeholders. Individual chapters will be released and marked complete as notated in the **revision history table below**. Chapters with content that has not been marked as released in the Revision History table should not be assumed to be complete nor reviewed.*

Who is this handbook for?

The handbook is intended for:

- engineers designing or specifying industrial vision systems;
- automation and controls engineers integrating vision equipment;
- computer-vision and machine-learning practitioners working in industry;
- students and lecturers studying machine vision;
- quality, validation and regulatory professionals;
- technical reviewers evaluating machine-vision projects.

The material is organised so that readers can use it either as a sequential engineering reference or as a source for individual design, implementation and validation topics.

Revision history

This table records formal content releases of individual sections. A section remains a placeholder until substantive content has been technically reviewed and approved for publication.

Revision	Date	Scope	Status	Notes
Develop- ment skeleton	2026-08-04	Handbook structure	Under development	Initial release of handbook structure

Acknowledgements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Notation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part I.

Foundations

i **Part status:** Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

2. Vision engineering: scope, definitions and history

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

2.1. Machine vision, computer vision and image processing

2.1.1. Working definitions and industrial success criteria

! Authoring syntax demonstration

This subsection is a deliberately fictional authoring example. It demonstrates Quarto syntax and project conventions; it is not approved handbook content and must not be cited as technical guidance.

2.1.1.1. Paragraphs, emphasis and terminology

For this fictional example, **machine vision** is treated as an engineered system that acquires image data, extracts task-relevant information and produces an operational decision. The distinction between *image quality*

and **inspection performance** is important: an aesthetically pleasing image is not necessarily the most reliable input for an industrial inspection.

The example uses the term **inspection score** for a dimensionless software output and the symbol p_{accept} for an estimated probability of acceptance. The notation x^2 demonstrates a superscript, while I_0 demonstrates a subscript.

The public [Quarto documentation](#) is an example of an external hyperlink.

A footnote may contain a clarification that would interrupt the main argument.¹

2.1.1.2. Bullet lists

A first-level list can describe broad engineering criteria:

- the feature must be visible;
- the measurement must be repeatable;
- the decision must fit within the required cycle time.

A nested list can distinguish criteria from supporting checks:

- Imaging performance
 - sufficient contrast at the feature boundary;
 - acceptable signal-to-noise ratio;
 - controlled geometric distortion.
- Decision performance
 - acceptable false-accept risk;
 - acceptable false-reject risk;
 - traceable pass/fail output.

A numbered workflow can express sequence:

¹This is a demonstration footnote. It is not an approved technical definition.

1. Define the inspection requirement.
2. Characterise the relevant feature.
3. Select illumination and optics.
4. Acquire representative images.
5. validate the complete decision chain.

2.1.1.3. Equations and equation references

An inline equation can be included within a sentence. For example, the object-space sampling interval is $s = W/N$, where W is the field width and N is the number of pixels across that width.

A displayed and numbered equation can represent an illustrative contrast measure:

$$C = \frac{I_{\text{feature}} - I_{\text{background}}}{I_{\text{feature}} + I_{\text{background}}} \quad (2.1)$$

The value calculated using Equation 2.1 is dimensionless. This equation is included only to demonstrate Quarto equation syntax and cross-referencing.

2.1.1.4. Figure and figure reference

The simplified fictional inspection chain is shown in Figure 2.1.

2.1.1.5. Table and table reference

Table 2.1 shows a small illustrative table. The values are fictional and must not be interpreted as engineering recommendations.



Figure 2.1.: Simplified block diagram showing illumination, optics, sensor, processing and decision stages connected from left to right.

Table 2.1.: Fictional industrial success criteria used to demonstrate a numbered Quarto table.

Criterion	Illustrative target	Evidence
Feature contrast	$C > 0.25$	Image study
Repeatability	$\sigma < 0.10$ mm	Repeatability trial
Cycle time	$t < 100$ ms	Timed production test

2.1.1.6. Code block and code-listing reference

A fenced code block can display source code without executing it. Listing 2.1 demonstrates a short Python example.

Listing 2.1 Fictional threshold decision used to demonstrate a numbered code listing.

```

1 def classify_measurement(value: float, limit: float) -> bool:
2     """Return True when the fictional measurement passes."""
3     return value <= limit
  
```

2.1.1.7. Callouts

i Note

A note callout can provide useful supporting information that is not central to the main argument.

 Authoring tip

Use semantic identifiers rather than identifiers derived from chapter or section numbers.

 Warning

The numerical values in this demonstration are fictional. They must not be copied into a real inspection specification.

2.1.1.8. Definition and example blocks

Definition 2.1 (Industrial success criterion). For this fictional example, an industrial success criterion is a measurable condition used to determine whether a complete vision system is suitable for its intended operational task.

The fictional definition in Definition 2.1 can be followed by a worked example.

Example 2.1 (Illustrative acceptance test). A test set contains 200 representative parts. The system must process every part within the specified cycle time and record the decision, confidence value and image identifier.

2.1.1.9. Citations and bibliography references

A parenthetical citation uses square brackets, for example (Szeliski 2022). Multiple works can be cited together (Szeliski 2022; Batchelor 2012).

A textual citation can be written as Szeliski (2022), while a citation with a locator can be written as (see Batchelor 2012, chap. 1).

2.1.1.10. Section and cross-chapter references

A reference to another section in this chapter can use Section [2.2](#).

A reference to a section in another chapter can use Section [3.1](#).

Cross-references should be preferred to manually written phrases such as “see Section 2.1”, because Quarto updates the displayed number and hyperlink when the structure changes.

2.1.1.11. Downloadable supporting file

The fictional dataset used for this example can be downloaded as [example inspection results](#)`{download}`.

2.1.1.12. Quotation

A useful inspection image is one that supports a reliable decision.

— Fictional Machine Vision Academy example

2.1.2. The four fields: image processing, computer vision, machine vision, AI

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.1.3. Relations, overlaps and typical confusions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.2. Historical development

2.2.1. Signal processing and the first digital images (1960s–1970s)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.2.2. Early computer vision and geometric approaches (1970s–1980s)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.2.3. The industrial era: machine vision on the factory floor (1980s–2000s)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.2.4. Statistical learning and the deep-learning turn (2000s–2010s)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.2.5. The regulatory era (2020s onwards)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.3. Industrial application families

2.3.1. Inspection and defect detection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.3.2. Guidance and robot vision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.3.3. Measurement and non-contact gauging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.3.4. Identification: codes, characters and traceability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.3.5. Emerging areas: biometrics, autonomy, medical imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.4. System architectures

2.4.1. Embedded and edge vision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.4.2. Smart cameras

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.4.3. PC-based and industrial-server systems

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.4.4. Cloud and SaaS vision platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.4.5. Hybrid and tiered architectures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.4.6. Comparison and selection criteria

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.5. The vision system as an engineered chain

2.5.1. The five stages: illumination, optics, sensor, processing, decision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.5.2. Constraint-first engineering: solving with light before software

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.5.3. How the chain interacts: propagating a requirement end to end

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.5.4. Professional roles and certification landscape

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.6. Approaching a vision problem: a design framework

2.6.1. From inspection requirement to imaging specification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.6.2. Characterising the feature: contrast, scale, geometry, tolerance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.6.3. Detection versus measurement: how the task drives every later choice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

2.6.4. A worked decision path, cross-referenced to later chapters



Content status: Placeholder.

This subsection has not yet been drafted or technically reviewed.

3. Light, colour and radiometry

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

3.1. Physical foundations of light

3.1.1. Electromagnetic spectrum and the visible band

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.1.2. Radiometric quantities: radiant flux, irradiance, radiance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.1.3. Photometric quantities and the photopic response

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.1.4. The inverse-square law and cosine law

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.2. Interaction of light with matter

3.2.1. Specular and diffuse reflection; the BRDF

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.2.2. Transmission, refraction and Snell's law

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.2.3. Absorption and fluorescence

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.2.4. Scattering and translucency

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.2.5. Implications for surface inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3. Colour

3.3.1. Trichromacy and the human visual system

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.2. Colour matching and the CIE XYZ system

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.3. Working colour spaces: RGB, sRGB, HSV, HSL, Lab

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.4. Colour temperature and illuminants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.3.5. White balance and colour constancy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4. Polarisation

3.4.1. States of polarisation and Stokes parameters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4.2. Polarisers, retarders and the Fresnel equations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4.3. Polarisation-resolving sensors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

3.4.4. Industrial uses: glare suppression, stress imaging, transparent-material inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4. Optics and lens systems

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

4.1. Geometric optics

4.1.1. The pinhole model

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.2. Thin- and thick-lens models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.3. Focal length, f-number, aperture

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.4. Field of view and working distance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.1.5. Depth of field and hyperfocal distance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2. Real lens behaviour

4.2.1. Spherical and chromatic aberration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.2. Geometric distortion: radial and tangential

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.3. Vignetting and cos fall-off

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.4. Modulation transfer function and resolving power

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.2.5. Diffraction limit

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3. Specialty optics

4.3.1. Telecentric lenses for metrology

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3.2. Macro and micro imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3.3. Line-scan and Scheimpflug configurations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.3.4. Liquid and variable-focus lenses

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4. Optical filters

4.4.1. Bandpass, longpass and shortpass filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.2. Polarising filters in imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.3. Neutral density, UV cut and IR cut filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.4. Interference filters and dichroic elements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.4.5. Filter placement and stacking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5. Selecting an optical configuration

4.5.1. Matching focal length to sensor size, FOV and working distance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.2. Resolution and DOF budgeting from the feature specification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.3. When telecentricity, filtering or specialty optics are required

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.4. Mount standards and mechanical envelope: C, CS, F, M42, S-mount

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

4.5.5. A lens-selection workflow, cross-referenced to Chapters 2 and 5

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part II.

Image Formation

i **Part status:** Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

5. Illumination for vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

5.1. Illumination principles

5.1.1. Light and geometry as the first solution

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.1.2. Contrast, feature enhancement and defect visibility

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.1.3. The lighting-first design method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2. Illumination geometries

5.2.1. Frontal and diffuse illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.2. Directional and grazing illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.3. Backlighting for silhouettes and dimensioning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.4. Coaxial (on-axis) illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.5. Dome and cloudy-day illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.6. Dark-field illumination

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.2.7. Structured light

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3. Light sources

5.3.1. LEDs: colour, drive, lifetime, thermal management

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3.2. Fluorescent, halogen and xenon sources

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3.3. Laser sources and safety classes

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.3.4. Continuous, strobed and pulsed operation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4. Wavelength selection

5.4.1. Visible-band choices and colour separation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.2. Near-infrared imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.3. Ultraviolet imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.4. Short-wave infrared and thermal imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.4.5. Multispectral and hyperspectral approaches

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.5. Environmental integration

5.5.1. Ambient light rejection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.5.2. Enclosures, shrouds and shielding

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.5.3. Vibration, contamination and thermal drift

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6. Designing an illumination solution

5.6.1. Characterising the feature and the surface

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.2. Single-plane versus multi-plane and 3D features

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.3. Edge visibility for measurement versus contrast for detection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.4. Matching geometry to task: a decision guide

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.5. Combining geometry, wavelength and filtering

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

5.6.6. Worked examples: from part to lighting choice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6. Image sensors and cameras

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

6.1. Sensor technologies

6.1.1. CCD architecture and readout

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.2. CMOS architecture: global vs rolling shutter

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.3. Line-scan and TDI sensors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.4. SPAD, event-based and neuromorphic sensors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.1.5. Multispectral and hyperspectral sensor architectures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2. Sensor performance and noise

6.2.1. Quantum efficiency and spectral response

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.2. Read noise, shot noise, dark current, fixed-pattern noise

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.3. Dynamic range and bit depth

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.4. Full-well capacity and saturation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.2.5. Signal-to-noise budgeting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3. Sampling, resolution and colour acquisition

6.3.1. Pixel pitch, sensor size and effective resolution

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3.2. Bayer patterns and demosaicing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3.3. 3-CCD and multi-sensor colour

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.3.4. Monochrome versus colour trade-offs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4. Camera interfaces and image transport

6.4.1. Interface selection criteria: bandwidth, cable length, CPU load, cost

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.2. USB3 Vision: bandwidth, cable limits and use cases

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.3. Camera Link, CoaXPress and frame-grabber-based interfaces

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.4. Legacy and niche interfaces: FireWire, analogue, MIPI CSI

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.4.5. Triggering, exposure control and time stamping

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5. GigE Vision in depth

6.5.1. Generational bandwidth: 1, 5, 10 and 25 GbE and practical reach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.2. The GigE Vision protocol: discovery, stream and control channels

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.3. Network configuration: jumbo packets, dedicated NICs, receive buffers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.4. Multi-camera topologies, switches and bandwidth budgeting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.5. Power over Ethernet and cabling

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.5.6. GenICam and cross-vendor interoperability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6. Selecting a camera and interface

6.6.1. From resolution and speed requirement to sensor choice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6.2. Matching interface to bandwidth, distance and architecture

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6.3. Area-scan versus line-scan decision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

6.6.4. A camera-and-interface selection workflow

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part III.

Image Processing

i **Part status:** Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

7. Digital images: representation and pixel-level processing

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

7.1. The digital image as a signal

7.1.1. Sampling, quantisation and the resulting data model

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.1.2. Coordinate systems, indexing and boundary conventions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.1.3. Colour, greyscale and multi-channel representations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.1.4. File formats and lossless vs lossy compression

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2. Point operations

7.2.1. Brightness, contrast and gain

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2.2. Look-up tables, gamma and tone curves

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2.3. Histogram equalisation and adaptive stretching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.2.4. Thresholding as a point operation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3. Arithmetic and logical operations

7.3.1. Pixel-wise arithmetic and blending

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3.2. Absolute and signed differences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3.3. Masking and region-of-interest operations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.3.4. Bitwise operations on binary images

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.4. Noise models and denoising choices

7.4.1. Gaussian, Poisson and impulse noise models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.4.2. Estimating noise from image statistics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

7.4.3. Choosing a denoiser: signal preservation vs artefact suppression

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8. Neighbourhood operations and image filtering

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

8.1. Convolution and correlation

8.1.1. Discrete convolution in two dimensions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.1.2. Separable and rank-one kernels

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.1.3. Boundary handling: zero, reflect, replicate, wrap

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.1.4. Computational cost and efficient implementations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2. Smoothing filters

8.2.1. Box (mean) filter and integral images

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.2. Gaussian filter and selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.3. Median and rank-order filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.4. Bilateral and edge-preserving filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.2.5. Non-local means

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3. Derivative and edge filters

8.3.1. First-order operators: Sobel, Prewitt, Scharr

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3.2. Second-order operators: Laplacian and Laplacian of Gaussian

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3.3. Gradient magnitude and orientation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.3.4. Sub-pixel edge localisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4. The frequency domain

8.4.1. The 2D discrete Fourier transform

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.2. Magnitude, phase and their perceptual roles

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.3. The convolution theorem

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.4. Low-pass, high-pass and band-pass filtering

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.5. Frequency-domain inspection of periodic structure

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.4.6. The sampling theorem and aliasing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5. Scale space and image pyramids

8.5.1. Scale-space theory in outline

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5.2. The Gaussian pyramid

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5.3. The Laplacian pyramid

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.5.4. Multi-resolution processing patterns

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6. Choosing a filtering strategy

8.6.1. Matching a filter to the noise model and the feature

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6.2. Preserving edges while suppressing noise

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6.3. Spatial versus frequency-domain: when each wins

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

8.6.4. Ordering operations in a preprocessing pipeline

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9. Segmentation, morphology and blob analysis

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

9.1. Thresholding methods

9.1.1. Global thresholds and manual selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.1.2. Otsu's method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.1.3. Adaptive and locally-adaptive thresholding

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.1.4. Multi-level and hysteresis thresholding

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2. Region-based segmentation

9.2.1. Region growing and merging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2.2. Watershed segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2.3. Superpixels: SLIC and successors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.2.4. Graph-cut segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.3. Edge-based segmentation

9.3.1. The Canny detector

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.3.2. Edge linking and boundary tracing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.3.3. Active contours (snakes) and level sets

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4. Mathematical morphology

9.4.1. Structuring elements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.2. Erosion and dilation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.3. Opening and closing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.4. Hit-or-miss transform and skeletonisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.5. Distance transforms and reconstruction

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.4.6. Greyscale morphology

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5. Connected-components and blob analysis

9.5.1. Connectivity and labelling algorithms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.2. Region properties: area, perimeter, centroid

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.3. Geometric moments and Hu invariants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.4. Shape descriptors: circularity, eccentricity, compactness

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.5.5. Filtering blobs by property

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.6. Choosing a segmentation approach

9.6.1. Thresholding versus region versus edge versus learned

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.6.2. Reliability under illumination and contrast variation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

9.6.3. Combining segmentation with morphological clean-up

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10. Feature detection, description and matching

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

10.1. Interest-point detection

10.1.1. The Moravec detector and the corner concept

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.2. Harris and Shi–Tomasi corners

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.3. Scale-invariant detection: difference of Gaussians

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.4. Hessian-based detectors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.1.5. FAST and AGAST corners for real-time use

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2. Feature description

10.2.1. SIFT descriptor

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2.2. SURF descriptor

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2.3. Binary descriptors: BRIEF, ORB, BRISK, FREAK

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.2.4. Learned descriptors: SuperPoint and successors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3. Feature matching

10.3.1. Distance metrics: L2, Hamming

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3.2. Brute-force and FLANN matching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3.3. The Lowe ratio test and cross-check

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.3.4. Robust matching with RANSAC and MSAC

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4. Template and pattern matching

10.4.1. Sum of squared and absolute differences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4.2. Normalised cross-correlation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4.3. Rotation- and scale-invariant template matching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.4.4. Geometric pattern matching (edge-model methods)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5. Line, circle and shape detection

10.5.1. The Hough transform for lines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.2. The Hough transform for circles

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.3. The generalised Hough transform

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.4. Direct fitting: least-squares lines, circles, ellipses

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.5.5. Robust fitting with RANSAC

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.6. Choosing a feature method

10.6.1. Repeatability, invariance and speed trade-offs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.6.2. Hand-crafted versus learned features

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

10.6.3. Matching the detector–descriptor pair to the task

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11. Geometric transformations, calibration and registration

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

11.1. Coordinate systems and homogeneous coordinates

11.1.1. Image, camera, world and object frames

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.1.2. Homogeneous representation of points and lines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.1.3. Matrix representation of transformations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2. The 2D transformation hierarchy

11.2.1. Rigid, similarity and affine transformations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2.2. Projective transformations (homographies)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2.3. Estimation from point correspondences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.2.4. Robust estimation of homographies

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.3. Image warping and interpolation

11.3.1. Forward vs inverse mapping

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.3.2. Nearest-neighbour, bilinear and bicubic interpolation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.3.3. Anti-aliased warping

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.4. The pinhole camera model

11.4.1. Intrinsic parameters and the camera matrix

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.4.2. Extrinsic parameters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.4.3. Radial and tangential distortion models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5. Camera calibration

11.5.1. Zhang's method and the planar-target approach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.2. Calibration targets and image-acquisition strategy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.3. Reprojection error and calibration diagnostics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.4. Uncertainty and re-calibration policy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.5.5. Hand-eye calibration for robot vision

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6. Image registration

11.6.1. Feature-based registration pipelines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6.2. Intensity-based (direct) registration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6.3. Registration for inspection: golden templates

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.6.4. Mosaicing and stitching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7. Image-comparison metrics

11.7.1. Absolute and squared differences; PSNR

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7.2. SSIM and multi-scale SSIM

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7.3. Histogram-distance metrics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

11.7.4. Learned perceptual metrics (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part IV.

3D Vision and Motion

i **Part status:** Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

12. Stereo and 3D vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

12.1. Two-view geometry

12.1.1. The epipolar constraint

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.1.2. The essential and fundamental matrices

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.1.3. Estimation from correspondences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.1.4. Stereo rectification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2. Stereo matching

12.2.1. Block-matching costs: SAD, SSD, NCC, census

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.2. Aggregation and disparity selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.3. Semi-global matching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.4. Learned stereo matchers (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.2.5. Disparity to depth and triangulation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.3. Uncertainty in 3D measurement

12.3.1. Depth-error propagation from disparity

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.3.2. Task uncertainty versus datasheet accuracy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.3.3. Uncertainty budgets for 3D measurement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4. 3D imaging modalities

12.4.1. Passive stereo

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.2. Structured-light triangulation and fringe projection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.3. Time-of-flight cameras

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.4. LiDAR and long-range sensing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.5. Photogrammetry, structure from motion and MVS

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.4.6. Comparison and selection guide

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5. Point clouds and 3D representations

12.5.1. Point-cloud data structures and file formats

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5.2. Filtering, downsampling and normal estimation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5.3. Meshing and surface reconstruction

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.5.4. Voxelisation and occupancy representations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.6. Point-cloud registration and alignment

12.6.1. Iterative closest point (ICP) and variants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.6.2. Feature-based 3D registration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.6.3. Global alignment strategies

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.7. 3D measurement in practice

12.7.1. Sensor acceptance testing with calibrated artefacts

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.7.2. Surface interaction: specular and translucent materials

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

12.7.3. Warm-up drift and environmental control

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13. Motion, tracking and event-based vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

13.1. Motion representation

13.1.1. The brightness-constancy assumption

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.1.2. Sparse vs dense motion fields

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2. Optical flow

13.2.1. The Lucas–Kanade method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2.2. The Horn–Schunck method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2.3. Dense flow: Farnebäck and pyramidal schemes

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.2.4. Learned optical flow (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.3. Background modelling and change detection

13.3.1. Frame differencing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.3.2. Gaussian mixture models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.3.3. Robust background subtraction

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4. Object tracking

13.4.1. Template and feature tracking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4.2. Mean-shift and correlation-filter trackers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4.3. Kalman filters and particle filters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.4.4. Multi-object tracking and data association

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5. Event-based vision

13.5.1. The event-camera imaging principle

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5.2. Representations: event frames, time surfaces, voxel grids

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5.3. Algorithms on event streams

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

13.5.4. Industrial fit and integration considerations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part V.

Deep Learning

i **Part status:** Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

14. Machine learning foundations for vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

14.1. The learning paradigm

14.1.1. Supervised, unsupervised and self-supervised learning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.1.2. Training, validation and test sets

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.1.3. Bias, variance and generalisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.1.4. Cross-validation and model selection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2. Classical classifiers

14.2.1. Linear and logistic regression

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.2. K-nearest neighbours

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.3. Support vector machines and kernels

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.4. Decision trees, random forests and boosting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.2.5. Bayesian classifiers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3. Feature engineering and dimensionality reduction

14.3.1. Hand-crafted image features

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3.2. Principal component analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3.3. Linear discriminant analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.3.4. Manifold learning (outline)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4. Loss functions and optimisation

14.4.1. Squared, absolute and Huber loss

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4.2. Cross-entropy and log-loss

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4.3. Gradient descent and its variants

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.4.4. Regularisation: L1, L2, elastic net

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5. Evaluation of classifiers

14.5.1. The confusion matrix

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5.2. Accuracy, precision, recall, F-score

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5.3. ROC and precision-recall curves; AUC

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.5.4. Class imbalance and industrial reality

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.6. Choosing classical versus deep approaches

14.6.1. Data availability and labelling cost

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.6.2. Interpretability and validation burden

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

14.6.3. When hand-crafted features still win

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15. Neural networks and deep learning

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

15.1. Feed-forward networks

15.1.1. Neurons, layers and activation functions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.1.2. ReLU, sigmoid, tanh, GELU

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.1.3. Softmax outputs and cross-entropy

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.1.4. Universal approximation and the role of depth

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.2. Backpropagation and automatic differentiation

15.2.1. The chain rule on the computational graph

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.2.2. Forward and reverse-mode differentiation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.2.3. Practical implementation in modern frameworks

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3. Optimisation of deep models

15.3.1. Stochastic gradient descent and mini-batching

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.2. Momentum, Nesterov and Adam

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.3. Learning-rate schedules and warm-up

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.4. Weight decay, dropout, batch and layer normalisation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.5. Data augmentation for images

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.3.6. Early stopping and validation monitoring

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4. Convolutional neural networks

15.4.1. The convolutional layer and parameter sharing

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4.2. Pooling, stride and receptive field

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4.3. Padding conventions and output-size arithmetic

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.4.4. Feature hierarchies and interpretability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5. Canonical architectures

15.5.1. LeNet-5, AlexNet, VGG

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.2. GoogLeNet / Inception

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.3. ResNet and residual connections

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.4. DenseNet, MobileNet, EfficientNet

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.5.5. Vision transformers and hybrid models

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6. Training strategy for industrial data

15.6.1. Transfer learning and fine-tuning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6.2. Small-data regimes and few-shot learning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6.3. Class-imbalance countermeasures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

15.6.4. Reproducibility: seeds, versioning, provenance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16. Deep learning for classification, detection, segmentation and anomaly

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

16.1. Image classification

16.1.1. Whole-image classification pipelines

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.1.2. Multi-label classification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.1.3. Fine-grained classification

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.1.4. Evaluation with imbalanced industrial datasets

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2. Object detection

16.2.1. Two-stage detectors: R-CNN, Fast R-CNN, Faster R-CNN

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.2. One-stage detectors: YOLO family, SSD, RetinaNet

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.3. Anchor-free detectors: FCOS, CenterNet

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.4. Transformer-based detection: DETR

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.2.5. Evaluation: IoU, mAP, per-class analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3. Semantic and instance segmentation

16.3.1. Fully convolutional networks

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.2. U-Net and encoder–decoder architectures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.3. DeepLab and dilated convolutions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.4. Mask R-CNN for instance segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.3.5. Panoptic segmentation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4. Anomaly and defect detection

16.4.1. Reconstruction-based methods: autoencoders and GANs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.2. Embedding-based methods: PatchCore, PaDiM, SPADE

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.3. One-class classifiers and Deep SVDD

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.4. Benchmarks: MVTec AD and industrial extensions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.4.5. Choosing between anomaly detection and supervised approaches

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5. Related tasks in industrial vision

16.5.1. Keypoint detection and pose estimation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5.2. Deep OCR and OCV

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5.3. Depth estimation from a single image

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

16.5.4. Self-supervised representation learning

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Part VI.

Deployment and Compliance

i **Part status:** Structural placeholder.
This part introduction will later summarise its scope, prerequisites and learning objectives.

17. From theory to tools: selection, cross-platform mapping and inspection-flow design

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

17.1. The industrial vision software landscape

17.1.1. Configurable tool sets versus code libraries

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.2. Open-source libraries: OpenCV, scikit-image, PyTorch/TF ecosystems

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.3. Commercial libraries and SDKs: Halcon, VisionPro, Sapura

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.4. Smart-sensor and no-code environments: Keyence, Zebra Aurora, In-Sight

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.1.5. Configuring versus programming: trade-offs and consequences

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2. Cross-platform tool mapping

17.2.1. How to read the mapping tables: theory concept to platform tool

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.2. Acquisition and preprocessing across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.3. Edge, line and circle detection: OpenCV / VisionPro / Halcon / In-Sight / Keyence / Zebra Aurora

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.4. Pattern location and alignment across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.5. Measurement and gauging tools across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.6. Defect and surface-inspection tools across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.7. Identification: barcode, Data Matrix, OCR/OCV across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.8. Deep-learning tools across platforms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.2.9. Limits of equivalence: where tools do not map cleanly

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3. Selecting and prioritising tools

17.3.1. From feature and task to candidate tools

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3.2. Prioritisation: robustness, speed, determinism, maintainability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3.3. Rule-based versus learned tools for a given problem

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.3.4. A tool-selection decision guide, cross-referenced to theory chapters

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4. Designing an inspection as a sequence of steps

17.4.1. Decomposing an objective into ordered steps

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4.2. The step pattern: objective, tools, region, output

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4.3. Locate–mask–measure–decide as a canonical flow

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.4.4. Worked example: scratch inspection decomposed into steps

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5. Thresholds, checks and pass/fail logic

17.5.1. Per-step checks: presence, contrast, dimension, position limits

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5.2. Setting limits and thresholds on tool outputs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5.3. Building the overall pass/fail decision from step results

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.5.4. Failure attribution: knowing which step failed and why

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6. Exposing KPIs and data-driven threshold setting

17.6.1. Choosing KPIs and measured outputs per tool and step

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.2. Logging tool outputs for offline analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.3. Batch statistical analysis to set thresholds and margins

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.4. Separating good/bad populations and setting operating points

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.6.5. Linking threshold setting to the validation of Chapter 18

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7. Software design practice for vision applications

17.7.1. Modularity, reuse and configuration management

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.2. Robustness: handling absent, occluded and out-of-spec parts

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.3. Defensive checks, error states and safe defaults

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.4. Version control, documentation and change control of vision jobs

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

17.7.5. Performance, cycle time and resource budgeting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18. Systems integration and industrial interfacing

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

18.1. Industrial communication

18.1.1. EtherNet/IP and PROFINET

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.1.2. Modbus TCP and serial protocols

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.1.3. OPC-UA and information modelling

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.1.4. MQTT and vision-to-cloud patterns

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.2. Integration with PLCs

18.2.1. Handshaking and trigger management

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.2.2. Result data structures and tags

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.2.3. Timing budgets and cycle time

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3. Integration with robots

18.3.1. Coordinate-frame transformations and hand-eye calibration

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3.2. Fixed and moving cameras

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3.3. Pick-and-place and vision-guided assembly

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.3.4. Bin picking and 3D-guided robotics

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4. Line integration

18.4.1. Reject mechanisms and part tracking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4.2. Statistical process-control feedback

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4.3. Traceability and data-logging architecture

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

18.4.4. Human–machine interfaces for vision stations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19. Validation, measurement uncertainty and sampling for inspection

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

19.1. Defining acceptance

19.1.1. False accept and false reject rates

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.1.2. The confusion matrix in industrial terms

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.1.3. Producer and consumer risk

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.1.4. Setting acceptance criteria from requirements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2. Repeatability and reproducibility

19.2.1. Sources of variation in a vision measurement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.2. Attribute Gauge R&R: method and calculation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.3. Variable Gauge R&R by the ANOVA method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.4. Kappa statistics and inter-rater/inter-system agreement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.2.5. Interpreting %R&R, %tolerance and number of distinct categories

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3. Confidence, tolerance and statistical intervals

19.3.1. Confidence intervals for proportions and means

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3.2. Tolerance intervals and coverage

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3.3. Confidence–reliability statements (e.g. 95/95)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.3.4. Propagation of uncertainty through a measurement chain

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4. Sampling plans for inspection validation

19.4.1. Attributes versus variables sampling

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.2. Binomial and hypergeometric bases for attribute plans

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.3. Operating-characteristic curves, AQL and RQL/LTPD

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.4. Zero-acceptance ($c=0$) plans and their use

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.5. Confidence–reliability sample-size formulae

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.4.6. ANSI/ASQ Z1.4, Z1.9 and ISO 2859 in practice

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.5. Measurement-system analysis for vision

19.5.1. Bias, linearity and stability studies

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.5.2. Uncertainty budgets and traceability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.5.3. Total measurement uncertainty for vision gauges

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6. Verification, troubleshooting and re-validation

19.6.1. A systematic troubleshooting method

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6.2. Isolating optical, lighting, sensor and software failures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6.3. Recording and root-cause analysis

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

19.6.4. Change control and re-validation triggers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20. Deployment, monitoring and human-in-the-loop for learning-based vision

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

20.1. From trained model to production system

20.1.1. Model export and interchange (ONNX)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.1.2. Quantisation, pruning and operator fusion

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.1.3. Runtime targets: CPU, GPU, edge accelerators

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.1.4. Revalidation of the deployed artefact

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2. Dataset governance

20.2.1. Collection planning against the operating envelope

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2.2. Labelling protocols and inter-annotator agreement

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2.3. Curation, exclusion records and provenance

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.2.4. Versioning and model-to-data traceability

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.3. Post-deployment monitoring

20.3.1. Domain shift and concept drift

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.3.2. Statistical drift detectors

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.3.3. Retraining loops and change control

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4. Human-in-the-loop systems

20.4.1. Why humans remain in critical and deep-learning loops

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.2. Review triggers: low confidence, drift, novelty, borderline cases

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.3. Interface and workflow design for human review

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.4. Human-in-the-loop as a labelling and improvement engine

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.5. Levels of autonomy and human oversight

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.4.6. Automation bias and the limits of human checking

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5. Explainability and interpretability

20.5.1. Saliency and Grad-CAM

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.2. LIME and local surrogates

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.3. SHAP and Shapley attributions

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.4. Limits of post-hoc explanation

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.5.5. Explanation as debugging versus explanation for audit

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6. Bias and fairness in industrial vision

20.6.1. Subgroups in industrial data: cavities, suppliers, shifts

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6.2. Subgroup analysis and reporting

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6.3. Mitigations: data, loss, imaging

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

20.6.4. Fairness when the system observes people

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21. Process validation and regulation of vision inspections in regulated industries

i **Chapter status:** Structural placeholder.
This chapter currently contains the approved heading structure but no reviewed technical content.

21.1. Process validation of inspection systems

21.1.1. IQ, OQ and PQ for vision systems

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.2. Validating an inspection against its acceptance criteria

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.3. Linking validation to the sampling and R&R methods of Chapter 18

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.4. Software validation and computer-system validation (CSV/CSA)

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.1.5. Validating learning-based inspections: special considerations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2. The EU Artificial Intelligence Act

21.2.1. Scope, definitions and territorial reach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.2. Prohibited practices

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.3. The high-risk category and Annex III

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.4. Obligations for providers and deployers of high-risk AI

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.5. Limited- and minimal-risk tiers

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.2.6. Timelines, enforcement and penalties

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3. US regulation for AI and inspection

21.3.1. The FDA framework for AI/ML-enabled devices

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.2. Predetermined change-control plans and the total-product-lifecycle approach

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.3. 21 CFR Part 11: electronic records and signatures

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.4. NIST AI Risk Management Framework

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.3.5. Comparison of EU and US approaches

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4. Medical-device and pharmaceutical inspection

21.4.1. MDR and IVDR requirements relevant to vision inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4.2. FDA 21 CFR 820 / QSR and process validation expectations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4.3. GMP and pharma visual-inspection requirements

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.4.4. Validating automated visual inspection against manual inspection

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.5. Other high-risk and safety-critical sectors

21.5.1. Aerospace inspection and qualification expectations

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.5.2. Automotive quality and functional-safety context

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

21.5.3. Common threads across regulated industries

i **Content status:** Placeholder.
This subsection has not yet been drafted or technically reviewed.

Bibliography

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